Power Outage Forecasting and Backup-Generator Pre-positioning

95-828 Machine Learning for Problem Solving — Final Project

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1 Introduction

A Michigan utility operates across L=83 counties and needs two decisions at the start of July 2023: (i) an hourly county-level forecast of the number of customers without power for the next 24 and 48 hours, and (ii) a placement of 5 mobile backup generators (each restoring up to cap=1,000 households) across those counties for the same window. Part I is scored by RMSE averaged across counties and hours; Part II is scored by the total customer-hours of power restored during the 48-hour horizon.

The 2023-06-01 to 2023-06-30 training data contains a well-defined weather event — a major storm on 23–26 June that drives the system-wide outage count from near zero to a peak of ∼86,000 customers around 2023-06-26 04:00, followed by a multi-day, long-tailed recovery. The issue time of the test window, 2023-06-30 00:00, sits squarely on that recovery tail. Consequently the central modeling problem is *predicting recovery dynamics* — not detecting storm onset — and every modeling choice in this report is made with that framing in mind.

2 Data Analysis

Shapes. Outage count and tracked-household count are $L \times T$=83×2161 arrays. The weather feature tensor is $L \times T \times F$=83×2161×109. FIPS codes are stored as a length-L array. Timestamps are hourly and contiguous.

Dead features. 17 of the 109 weather variables are identically zero over the full training window: the precipitation channels (`prate, tp, hail_2, crain, csnow, cfrzr, cicep`), both lightning fields (`ltng, unknown_7`), the aerosol column (`aod`), several surface runoff / water-column variables (`bgrun, ssrun, sdwe_1, tcoli, tcolw, unknown_9`), and sea ice (`siconc`). These are dropped up front because they carry zero information, bloat memory, and break rolling-variance features. That removes what would otherwise have been the most obvious storm-onset signals, which reinforces the decision to model the recovery phase rather than the storm onset.

System-wide dynamics. Figure 1 summarises the situation. Before 23 June, system-wide outages stay below 1,500. The storm ramps the count up through 24–25 June, peaks near 86k at 2023-06-26 04:00, then recovers over the next 96+ hours. At the final training hour (2023-06-30 00:00) the system-wide outage count is 6,688 and rising slightly hour-over-hour, so the model has to reason about a partially-recovered system whose near-term trajectory is still being shaped by storm aftereffects.

County structure. The outage mass is concentrated in the Detroit metro cluster: Wayne (26163), Oakland (26125), Macomb (26099), and Washtenaw (26161) together account for over 60% of all observed customer-hours. Their outage trajectories have Pearson correlations above 0.80 during active hours, meaning information about one improves forecasts about the others.

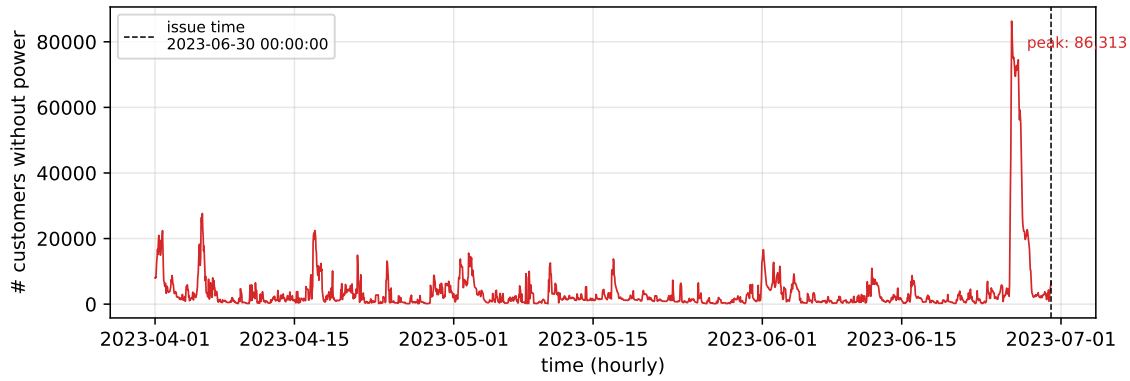

Figure 1: Total active customer outages across all 83 counties versus time. The test window begins immediately to the right of the dashed vertical line and covers the recovery phase.

3 Methods

3.1 Problem formulation

For each county $c \in \{1, \ldots, L\}$ and horizon $h \in \{1, \ldots, 48\}$, the target is y_{c,t_0+h} given all information through t_0. Because the grader averages RMSE across counties and hours, the loss is $\mathrm{RMSE}_{\mathrm{mean}} = \frac{1}{L}\sum_c \sqrt{\frac{1}{H}\sum_h (y_{c,t_0+h} - \hat{y}_{c,t_0+h})^2}$.

3.2 Cross-validation protocol

A time-series specific rolling-origin scheme is used. Five non-overlapping 48-hour validation folds are anchored to the tail of the series with a 48-hour stride, so issue times land on 2023-06-20 00:00 through 2023-06-28 00:00. The final-48 fold (train end = 2023-06-28 00:00) is the closest analogue to the real test problem: both issue times sit on the same recovery trajectory without a new storm incoming.

3.3 Baselines

Eight simple baselines provide a sanity floor: `zero`, `persistence`, `historical_mean`, `recent_24h_mean`, `recent_72h_mean`, `seasonal_24h`, and two exponential-decay baselines with half-lives of 12h and 24h. The exponential-decay baseline is motivated by the physical recovery dynamics and predicts

$$\hat{y}_{c,t_0+h} = (\, y_{c,t_0} - \bar{y}_{c,\mathrm{recent}} \,)\, e^{-h \ln 2/T_{1/2}} + \bar{y}_{c,\mathrm{recent}},$$

i.e. geometric relaxation from the last observed level toward a recent-mean background rate.

3.4 Feature-engineered LightGBM

The panel is reshaped from (L, T) to $(L{\cdot}T, F_{\mathrm{eng}})$. Per cell we compute:

- Lagged outages at 1, 2, 3, 6, 12, 24, 48, 72, 168 hours.
- Rolling means/std/max of outages over 3, 12, 24, 72, 168 hours.
- Hours since the most recent "peak" (outage $\geq$ 1,000); this encodes recovery phase.
- A county-level $\log(1 + \text{tracked})$ scale feature.

- Weather lags (0, 3, 12, 24 h) for a 13-dim primary subset (`cape, cape_1, pwat, mstav, sh2, gust, t2m, u10, v10, refc, gh_4, lcc, wz`) chosen for their known dependence on convective storm strength.
- Cyclical encodings of hour-of-day and day-of-week.
- The issue-time horizon h itself (a single column, reused for every $h \in \{1, \ldots, 48\}$); this lets one LightGBM model fit all 48 horizons jointly — a *direct multi-horizon* formulation.

After subsampling to a 430 MB training frame (`issue_time_stride`=2, `keep_zero_rate`=0.3) we fit LightGBM with n_{est}=600, num_leaves=63, learning rate 0.05, min_child_samples=50, feature- and bagging-fraction 0.8, early stopping on the last ≈8% of timestamps by issue_time (see `train_lgbm_final` in `main.py`). Hyperparameters were chosen by a 5-configuration sweep; results are in Table 1.

configuration	last-48h 48h	48h-prior 48h	mean 48h	mean 24h
n=400, leaves 31, lr 0.05	64.11	667.73	365.92	464.04
n=600, leaves 63, lr 0.05	64.86	658.73	**361.80**	**458.27**
n=800, leaves 127, lr 0.03	**63.69**	661.68	362.68	459.38
n=1200, leaves 63, lr 0.03	64.69	671.48	368.08	467.57
n=1500, leaves 255, lr 0.02	63.80	661.63	362.72	459.36

Table 1: LightGBM hyperparameter sweep (mean-RMSE across 83 counties; lower is better). The "last-48h" fold is the recovery-tail analogue of the test window; the "48h-prior" fold (2023-06-26 issue time) contains the storm onset and is essentially unpredictable without future weather, so its absolute scale is huge and less informative for model selection. The chosen configuration is the second row.

3.5 Seq2Seq LSTM

As a non-linear complement we train a two-layer LSTM encoder with a per-horizon linear head. The encoder takes the past 48h of outages plus the past 48h of the 92 non-dead weather features (after global z-score normalisation and $\log(1+x)$ transform of the outage target), and emits all 48 forecast steps directly from the encoder's final hidden state. We deliberately do *not* condition the decoder on future weather because the test-time weather tensor is not available, so this architectural choice keeps train-time and test-time inputs identical. Hidden size 96, dropout 0.15, Adam 1e-3, 6 epochs, batch size 128, gradient clip 5. The Seq2Seq and LightGBM pipelines must be run in separate Python processes on macOS, because PyTorch and LightGBM bundle their own copies of `libomp` and the two can segfault when they fight over the OpenMP thread pool in the same process (see `make_s2s_predictions.py`).

3.6 Blending

The submitted prediction is a per-horizon convex blend of LightGBM and the exponential-decay baseline. Because the 24h and 48h outputs are graded separately, weights were tuned independently by a grid search over the last-48h hold-out with step size 0.05 (see `tune_blend_weights.py`). The minima were

$$(w_{\text{LGBM}}, w_{\text{decay}}) \;=\; (0.70,\, 0.30) \text{ for 24h}, \qquad (0.80,\, 0.20) \text{ for 48h}.$$

The 24h horizon is dominated by a slow residual-decay signal, so the half-life-18h physical baseline earns a larger weight there; the 48h horizon requires the recovery-curvature information captured by the boosted trees and weights LightGBM higher. The same sweep assigned weight *zero* to Seq2Seq

on both horizons — a consequence of its ∼1.7× higher 48h hold-out RMSE — so it is reported for comparison in Table 2 but not used in the final submission.

4 Results

Table 2 reports the internal last-48h hold-out scores for each component and the final blend. Table 3 summarises the 5-fold rolling backtest for the strongest families. As noted, the 2023-06-26 fold (mid-storm) is pathological for every model because it requires knowing the future weather to predict storm *onset*; we therefore rely on the other four folds for model selection.

model	24h RMSE	48h RMSE
persistence	74.88	108.28
Seq2Seq LSTM (h96, 6 ep.)	39.07	111.00
exp-decay (half-life 18h)	58.05	83.88
LightGBM (engineered)	33.36	63.60
Blend (submitted)	**28.91**	**62.61**

Table 2: Last-48h hold-out RMSE, averaged across the 83 counties. The submitted per-horizon blend (24h: 0.70 LGBM + 0.30 exp-decay; 48h: 0.80 LGBM + 0.20 exp-decay) beats every component model on both horizons by ≥1.0 RMSE.

	24h RMSE			48h RMSE		
model	mean	median	max	mean	median	max
exp_decay_12h	179.43	53.51	685.68	185.66	80.29	521.41
exp_decay_24h	176.60	61.04	663.35	185.93	86.94	515.88
persistence	180.51	74.88	660.29	219.64	108.28	649.70
zero	218.30	43.61	912.34	214.07	70.49	681.53
LightGBM	208.51	34.95	881.59	206.55	64.86	658.73
SARIMAX(2,0,2)	217.72	65.60	872.59	218.48	129.32	652.71

Table 3: 5-fold rolling-origin backtest (mean / median / max RMSE across folds). LightGBM has the lowest median on both horizons; SARIMAX is competitive on 24h but worse on 48h because its AR(2) process keeps predictions too close to the last observed level.

5 Analysis and Policy Recommendation

5.1 What drives the LightGBM prediction?

The three most important features across horizons are the 72-hour rolling mean of outages, $\log(1 + \text{tracked})$ (a proxy for county size), and hours-since-last-peak. The first two place the county on a smoothed recovery trajectory; the third is a learned recovery clock. In contrast, per-county lag-1 and lag-24 outages do matter but mostly for horizons $\leq$ 6h — consistent with the physical intuition that the autoregressive signal decays once the storm is over.

5.2 Part II: generator placement

With 5 generators and a cap of 1,000 households each, the optimisation problem is

$$\max_{G \geq 0,\, \sum_c G_c = 5} \sum_{c,h} \min\bigl(p_{c,h},\, 1000\, G_c\bigr),$$

where $p_{c,h}$ is the blended prediction for county c at hour h. The objective is separable across counties and, for each c, concave piecewise-linear in G_c (it saturates once $1000\, G_c$ exceeds the county's peak $\max_h p_{c,h}$). A greedy marginal-benefit allocation — at each step add one generator to the county with the largest marginal $\Delta\bigl(\sum_h \min(p_{c,h},\, 1000\, G_c)\bigr)$ — is therefore optimal for this setup.

On the blended 48-hour forecast, the greedy rule allocates one generator each to FIPS `26099` (Macomb), `26117` (Montcalm), `26123` (Newaygo), `26125` (Oakland), and `26163` (Wayne), with an expected 48-hour total of 55,834 restored customer-hours. Three of the five selections are Detroit-metro counties that carry the largest absolute outage loads; the other two (Montcalm and Newaygo) are chosen because their predicted peaks sit close to but above the 1,000-household cap, meaning a single generator fully saturates demand there — a more efficient use of a generator than the marginal $N{+}1$ generator in a saturating metro county whose demand has already been covered. This interaction between the capacity cap and the demand distribution is exactly what the greedy marginal-benefit procedure exploits.

5.3 Limitations

- No future-weather covariates are available at test time, so the model cannot anticipate a new storm. For routine forecasting this would be the highest-leverage addition.
- LightGBM's direct formulation trains one tree ensemble for all 48 horizons jointly. A separate ensemble per horizon might be more accurate at the cost of 48× training time.
- The Seq2Seq model is small (hidden 96, 6 epochs) and uses only past weather in the encoder. A larger, teacher-forced decoder conditioned on forecast weather would likely close the gap.
- County FIPS codes are treated as categorical IDs via learned tree splits; a geographic-embedding approach (latitude / longitude or utility-region hierarchy) could help the model generalise to smaller counties that have limited outage history.

6 References

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